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Measuring Transparency in Consumer Contracts: The Usefulness of Readability Formulas Empirically Assessed

Article 5 of the Unfair Contract Terms Directive requires terms in consumer contracts to be ‘drafted in plain, intelligible language’. The European Court of Justice has ruled on several occasions that transparency essentially equates to ‘foreseeability’: if a term’s legal consequences can be foreseen by the average consumer, it is transparent. Foreseeability therefore requires, so we hypothesise, that the legal effects of terms are comprehensible. Terms are comprehensible if they are plain and intelligible, which means that they are readable. Courts struggle with this transparency test and some authors call for an empirical approach. We wonder whether empirically tested readability formulas could help legal professionals answer this call, albeit in a more efficient, less time-consuming way. Indeed, an obvious method to assess foreseeability would consist in the application of a readability formula. In this paper we focus on readability formulas, more specifically on the Flesch readability formula. We also devise our own formula, one that is specifically geared towards legalese. We then test both formulas – the Flesch formula and our own – empirically in order to evaluate their usefulness. Since we wish to address legal effects in a broad sense (without restricting ourselves to financial consequences) we opted to assess the foreseeability of general terms and conditions in gaming contracts. These terms and conditions have recently been criticised for being detrimental to the consumer (in the French Steam-case). Our main findings are threefold. Our first conclusion is that the Flesch readability formula cannot indicate whether terms are foreseeable. Second, our revised formula also failed the test. There is no correlation between readability as assessed by either formula and the actual comprehensibility of the terms. Third, we found that the legal effects of the terms and conditions in gaming contracts we presented to our participants were understood well and were thus foreseeable.

I. Introduction

Under European Union law, article 5 of the Unfair Contract Terms Directive (UCTD)¹ requires terms in consumer contracts to be ‘drafted in plain, intelligible language’.² This requirement is commonly referred to as *Transparenzgebot* or ‘transparency requirement’. What exactly constitutes plain and intelligible language has been subject to academic debate for centuries.³ Modern efforts can be linked back to the ‘plain English movement’ of the 1970s.⁴ Members of this movement used popular media to ridicule complicated legal language found in government documents, contracts, and more, demanding these texts to be drafted in a simpler language.⁵ This resulted in plain language legislation around the world, including the UCTD in the EU in 1993. That Directive, according to Cutts, ‘persuaded many previously stick-in-the-mud companies to write their contracts more plainly’.⁶

Requiring transparency in consumer contracts is one thing but evaluating if terms actually are transparent is another. The European Court of Justice (CJEU) has developed a line of case law on the transparency test laid down in article 5 UCTD. Most importantly, it ruled in *Invitel* that transparency essentially equates to ‘foreseeability’: if a term’s legal consequences can be foreseen by the consumer, it is transpar-

ent.⁷ Assessing foreseeability, however, is a complicated task. There does not exist a standard test to be performed by the national courts. We thus decided to take a closer look at methods that could be used to assess foreseeability.

To start with, the legal effects of a term can only be foreseen if the term is (made) available to the consumer. This touches upon the formal dimension of transparency.⁸ Formal transparency relates to the availability and accessibility of the terms and asks for an assessment of the active handing-over, font size and colour, placement on a website of contract terms, etc. Once the consumer disposes of the terms, substantive transparency comes into play. Substantive transparency, by some also referred to as material transparency,⁹ looks at the actual content of the terms, instead. Are the terms drafted in a way that the consumer is not misled about her rights? According to the CJEU, foreseeability first and foremost implies that a term’s consequences are *well understood* by the *average consumer*.¹⁰ Foreseeability thus requires, so we hypothesise, that terms are comprehensible. Terms are comprehensible if they are plain and intelligible,¹¹ which means that they are readable. An obvious method to assess foreseeability would then be the application of a readability formula.

In this paper we focus on substantive transparency and readability formulas. Our study tests the usefulness and effectiveness of these formulas to assess foreseeability, as an important element of transparency of terms and conditions. The most common readability formula is the *Flesch Reading Ease* formula.¹² In the eyes of Flesch’s readability formula, multisyllabic words and long sentences are less readable. Critics of this formula disagree with Flesch. Instead of focussing on syllables, attention should be paid to how common a word is and in which context it is used.¹³ To address this, we developed our own revised version of the Flesch readability for-

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1 Council Directive 93/13/EEC on unfair terms in consumer contracts [1993] OJ L95/29.

2 Article 5 UCTD.

3 M Cutts, *The plain English guide* (Oxford University Press 1995) 4 f.

4 M Shiflett, ‘Plain English Movement and Its Influence on Today’s Legal English’ (2017) 4 International Journal of Novel Research in Interdisciplinary Studies 11-14.

5 Cutts (n 3) 6 f.

6 *ibid* 7.

7 C-472/10 *Nemzeti Fogyasztóvédelmi Hatóság v Invitel Távközlési Zrt* [2012] ECLI:EU:C:2012:242, paras 24, 26, 28.

8 M Loos, ‘Transparency of standard terms under the Unfair Contract Terms Directive and the Proposal for a Common European Sales Law’ (2015) 23 European Review of Private Law 179, 180.

9 S Cámara Lapuente, ‘European Union: The Innovative Role of the CJEU’s Case-Law’ in YM Atamer and P Pichonnaz (eds), *Control of Price Related Terms in Standard Form Contracts* (Springer 2020) 77.

10 V Mak, ‘De ‘gemiddelde consument’: van fictie naar feit?’ (2017) 66 *Ars Aequi* 592, 595; C-26/13 *Árpád Kásler v OTP Jelzálogbank Zrt* [2014] ECLI:EU:C:2014:282, para 74.

11 See also: M Dellacasa, ‘Judicial review of “core terms” in consumer contracts: defining the limits’ (2015) 11 European Review of Contract Law 152, 171.

12 M Zamanian and P Heydari, ‘Readability of Texts: State of the Art’ (2012) 2 Theory and Practice in Language Studies 43, 44.

13 J Seely, *Oxford Guide to Effective Writing and Speaking: How to Communicate Clearly* (Oxford University Press 2013) 120 ff.

mula, a version geared towards legalese. Legalese is the formal and technical language of legal documents. We declared common words as ‘easy to read’ and legalese as ‘hard to read’.

Our study then tests both formulas empirically. We have presented our participants with terms and conditions that are commonly used in the gaming industry.¹⁴ Each participant was presented with three randomly selected terms. Each term was accompanied by a closed, multiple-choice question. All questions related to the consequences of the respective terms. For example, we presented participants with a choice of forum clause and asked them where they could bring a lawsuit. The results of this study allow us to draw preliminary conclusions as to whether readability formulas such as the formula developed by Flesch and our revised formula are indeed suitable methods to assess foreseeability.

The results also illustrate whether common terms and conditions in gaming contracts are well understood. Why did we opt for gaming contracts? Current literature (and case law) regarding transparency and readability heavily focusses on financial contracts and on financial effects of contracts, for example the price of a service or default interest rates.¹⁵ We see the need to look at all kinds of consequences, and not only those which are purely economic. Consumers now enter, on a daily basis, into contracts that are arguably very complicated, but that do not concern financial aspects at all. Consumers using digital distribution networks for computer or video games (‘gaming industry’) are confronted with increasingly restrictive terms and conditions.¹⁶ These contracts concern rights and obligations connected to material, but also immaterial wealth.¹⁷ Think of the progress of video games linked to a *Steam* account (‘safe games’). What if *Steam* suddenly stopped offering its services and the consumer lost access to her safe games? They will likely be worth nothing, materially speaking. Their immaterial value can be substantial, but chances are that the service’s terms and conditions have prophesied such loss and excluded any liability. Legal scholarship has focussed intensively on transparency in financial markets, but seldom on issues not involving clear-cut economic effects on the consumer.

Prior research has examined the transparency principle doctrinally and in parts also empirically, but empirical studies on whether readability formulas are suitable for assessing the foreseeability of a term’s consequences do not exist.¹⁸ The goal of this paper is to fill this gap in the transparency literature by offering empirical insights. Our motivation to do so stems from the movement calling for a more empirically informed consumer law.¹⁹ We ought out to answer the following question: *can readability formulas indicate whether terms and conditions in the gaming industry are transparent in the sense that their consequences are foreseeable?* To answer this question, Chapter III dives deeper into the transparency principle of article 5 UCTD and the court-developed foreseeability requirement. It also summarises different methods to test transparency, including readability formulas and our own version thereof. Chapter IV presents the methodology, collected data, and results of our empirical study. Lastly, Chapter V offers final remarks and a conclusion to the research question.

II. Transparency

The principle of transparency ensures that the consumer receives sufficient information from the commercial party.²⁰

Examples of the transparency requirement can be found in the Electricity Directive (2009/72/EC) and its replacement, the Directive on common rules for the internal market for electricity (EU/2019/944).²¹ Most prominently, article 5 UCTD stipulates that all contract terms must be drafted in ‘plain, intelligible language’.

1. Foreseeability and the UCTD

It was not before the *Invitel* judgement in 2012 that the CJEU started to offer deeper insights into the transparency principle of article 5 UCTD. The court was asked to comment on a term that allowed the commercial party, a telecom provider, to charge extra fees when payment was made by money order.²² However, that clause did not specify the actual amount of the fee; which was left to the scrutiny of the commercial party. The court made clear that a clause such as *in casu* must include ‘the reason for and the method of’ calculating the fee,²³ and that this information must be drafted in language ‘plain and intelligible to the consumer’.²⁴ Only then it would grant the consumer the possibility to foresee the amount of the fees.²⁵ This judgement therefore formally introduced a foreseeability requirement.²⁶

14 For a list of these terms, please consult the supplementary materials: <https://bit.ly/2ZofBhH>.

15 The broader perspective of VKI v Amazon being an exception: C-191/15 *Verein für Konsumenteninformation v Amazon EU Sàrl* [2016] ECLI:EU:C:2016:612.

16 TGI Paris, 4ème chambre, 17 September 2019; *UFC Que choisir c Valve Corp*, RG n°16/01008; This mainly relates to Digital Rights Management (DRM) and the move to online distribution. On DRM, see: N McDonald and others, ‘Digital Rights Management: The Four Perspectives of Developers, Distributors, Users, and Lawyers’ in SM Furnell and NL Clarke (eds), *Human Aspects of Information Security & Assurance* (Plymouth University 2015).

17 Immaterial wealth is one’s ‘own qualities and faculties for action and for enjoyment; such for instance (...) the faculty of deriving recreation from reading or music.’ Alfred Marshall, *Principles of Economics* (8th edn, Palgrave MacMillan 1920) chapter 3.

18 Van Boom et al looked at insurance policies. They studied whether readability scores have an influence on the expectations claimants have for reimbursement and whether lower expectations lead to a different behaviour in using the insurance (ask for advice, submit report, start legal proceedings): WH van Boom, P Desmet and MR van Dam, ‘Eenvoudige tekst, eenvoudige claim? Over de invloed van tekstniveau in algemene verzekeringvoorwaarden op verwachtingen en gedrag van consumenten’ (2015) *Tijdschrift voor Consumentenrecht en Handelspraktijken* 200, 205 f.

19 A-L Sibony, ‘Data and Arguments: Empirical Research in Consumer Law’ in H-W Micklitz, A-L Sibony and F Esposito (eds), *Research Methods in Consumer Law* (Edward Elgar 2018); P Møgelvang-Hansen, ‘Consumer Law and Consumer Behaviour: Some Illustrations’ in B Heiderhoff and R Schulze (eds), *Verbraucherrecht und Verbraucherverhalten* (Nomos 2016); G Van Dijk, S Sverdllov and G Buck, ‘Empirical Legal Research in Europe: Prevalence, Obstacles, and Interventions’ [2018] *Erasmus Law Review* 105; A-L Sibony, ‘Can EU Consumer Law Benefit from Behavioural Insights? An Analysis of the Unfair Practices Directive’ (2014) 22 *European Review of Private Law* 901-941.

20 Commission Notice on the Guidance on the interpretation and application of Council Directive 93/13/EEC of 5 April 1993 on unfair contract terms in consumer contracts [2019] 5325 final, 25 ff.

21 Article 28 (1) Council Directive (EU) 2019/944 on common rules for the internal market for electricity and amending Directive 2012/27/EU (recast) [2019] OJ L158/125; Council Directive 2009/72/EC concerning common rules for the internal market in electricity and repealing Directive 2003/54/EC [2009] OJ L211/55.

22 *Nemzeti Fogyasztóvédelmi Hatóság v Invitel Távközlési Zrt* (n 7) 18.

23 *ibid* 24, 26.

24 *ibid* 30, 31.

25 *ibid* 28; C-472/10 *A.G. Trstenjak Opinion in Nemzeti Fogyasztóvédelmi Hatóság v Invitel Távközlési Zrt*. [2011] ECLI:EU:C:2011:806, para 79.

26 C Leone, ‘Transparency revisited – on the role of information in the recent case-law of the CJEU’ (2014) 10 *European Review of Contract Law* 312, 319 f.

In *RWE Vertrieb* the court confirmed its earlier position and clarified that the mere possibility of obtaining information after the conclusion of the contract, be it initiated by the commercial party or not, does not suffice to ‘cure’ the lack of transparency of a contract term allowing for a price change.²⁷ Any obligation of other secondary law instruments to inform the other party before exercising the right laid down in the disputed term is complementary and cannot replace the test of transparency.²⁸ Such other secondary law would simply amount to a ‘*specially regulated example (...) of the transparency requirement already secured*’ under the UCTD.²⁹ The court also clarified that the principle of transparency does not only include formal transparency (access to information, contract of adhesion, etc.)³⁰ but also substantive transparency (possibility for a consumer to acknowledge and foresee her rights and duties under the contract).³¹

The notion of foreseeability of the consequences of a contractual term as established in *Invitel* and *RWE Vertrieb* was further specified in *Kásler*. In *Kásler*, the court assessed a contractual term that allowed the commercial party to unilaterally adjust monthly repayment instalments based on the exchange rate of a foreign currency.³² The court stated that the average ‘consumer (shall) foresee, on the basis of clear, intelligible criteria, the economic consequences for him’.³³ What is remarkable in *Kásler* is that the court seems to have narrowed its foreseeability test. While earlier case law did not refer to a particular kind of consequences, this case shows that *economic* consequences might be of particular importance.³⁴ The court has since reapplied this economic concept in *Matei*,³⁵ *Van Hove*,³⁶ *Bucura*,³⁷ *Banco Primus*,³⁸ *Andriuc*,³⁹ and *OTP Bank*.⁴⁰ Transparency and economic consequences are often interlinked, which is barely surprising given the financial illiteracy of the average consumer.⁴¹

This being said, in his *Kásler* opinion, Advocate-General Wahl, however, talked of ‘*advantages and disadvantages of concluding a certain contract and the risks* (associated to) *the transaction*’ while discussing the requirements of foreseeability.⁴² One could argue that these ‘risks’ include any advantages and disadvantages, regardless of their nature. That the court eventually referred to ‘*economic consequences*’ seems to merely have to do with the case at hand and must be seen as only one of many possible forms of consequences. The Grand Court has followed this view in the joined cases of *Gutiérrez Naranjo and Martínez* in 2016.⁴³ Instead of referring to more recent case law in *Kásler* and relying on the notion of economic consequences, it made the – arguably intentional – choice to reference *RWE* and its broader test of foreseeability of any ‘*financial or legal consequences*’ of the contract.⁴⁴

In *VKI v Amazon*, the court had already put the emphasis on the foreseeability of the *legal* consequences of a term, *in casu* the applicability of another law through an opaque choice of law clause.⁴⁵ Since these consequences were in breach of mandatory law, the term was deemed misleading and unfair in the sense of article 3 (1) UCTD.⁴⁶ In other cases the lack of foreseeability was not in itself decisive for the fairness test.

2. Assessing foreseeability

The introduction of the foreseeability requirement gave rise to two main questions. First, *who* must foresee the consequences? Second, how can foreseeability be assessed? This paper focusses on objective ways to do so. It briefly mentions

aspects of formal transparency but focusses on substantive transparency and how readable the terms are.

a. Average vs individual consumer

Who precisely must foresee the consequences and what must this person take into account? In *Kásler*, the Court – for the first time – established that not the individual, but rather the ‘average consumer’ should be set as the applicable standard.⁴⁷ EU internal market law has, in its case law, defined the ‘average consumer’ as ‘*reasonably well-informed and reasonably observant and circumspect*’.⁴⁸ However, it could be argued that this standard does not apply to the transparency principle as the court – in *Océano* and its later case law – sets out the goals of unfair terms legislation as protecting ‘*the consumer (...) in a weak position vis-à-vis the seller*’.⁴⁹ Applying a standard seen as too demanding by many would undermine that aim.⁵⁰ That said, the concept of the average consumer is sufficiently flexible to avoid setting the bar too high. We argue that the right approach is to look at the ‘average consumer’ in the particular field of business/services that the contract falls in.⁵¹ Questions that have to be answered are, though not exclusively, what is the current median age of consumers of the product, are those primarily female or male, and what is their average level of education. The average consumer is then this current average consumer with the only nuance that she might not yet have collected experience in the field of business at hand. For example, in the gaming industry, this would be an average computer and

27 C-92/11 *RWE Vertrieb AG v Verbraucherzentrale Nordrhein-Westfalen e. V.* [2013] ECLI:EU:C:2013:180, para 51.

28 *ibid* 52.

29 C-92/11 A.G. Trstenjak *Opinion in RWE Vertrieb AG v Verbraucherzentrale Nordrhein-Westfalen e. V.* [2012] ECLI:EU:C:2012:566, paras 68 f.

30 H-W Micklitz, ‘Unfair Terms in Consumer Contracts’ in N Reich and H-W Micklitz (eds), *European Consumer Law* (Intersentia 2014) 143.

31 *RWE Vertrieb AG v Verbraucherzentrale Nordrhein-Westfalen e. V.* (n 27) 49.

32 *Árpád Kásler v OTP Jelzálogbank Zrt* (n 10) 21.

33 *ibid* 73.

34 C-26/13 A.G. Wahl *Opinion in Árpád Kásler v OTP Jelzálogbank Zrt* [2014] ECLI:EU:C:2014:85, para 88.

35 C-186/16 *Ruxandra Paula Andriuc and Others v Banca Românească SA* [2017] ECLI:EU:C:2017:703, para 74.

36 C-96/14 *Jean-Claude Van Hove v CNP Assurances SA* [2015] ECLI:EU:C:2015:262, para 47.

37 C-348/14 *Maria Bucura v SC Bancpost SA* [2015] ECLI:EU:C:2015:447, para 54.

38 C-421/14 *Banco Primus SA v Jesús Gutiérrez García* [2017] ECLI:EU:C:2017:60, para 62.

39 *Ruxandra Paula Andriuc and Others v Banca Românească SA* (n 35) 50.

40 C-51/17 *OTP Bank v Teréz Ilyés* [2018] ECLI:EU:C:2018:750, para 78.

41 Vanessa Mak and J Braspenning, ‘Errare humanum est: Financial Literacy in European Consumer Credit Law’ (2012) 35 *Journal of Consumer Policy* 307-332.

42 A.G. Wahl *Opinion in Árpád Kásler v OTP Jelzálogbank Zrt* (n 34) 81.

43 Joined Cases C-154/15, C-307/15 and C-308/15 *Naranjo and Martínez* [2016] ECLI:EU:C:2016:980, para 20.

44 *ibid*, paras 20, 50.

45 *Verein für Konsumenteninformation v Amazon EU Sàrl* (n 15) 65 ff.

46 *ibid* 71.

47 *Árpád Kásler v OTP Jelzálogbank Zrt* (n 10) 74.

48 C-210/96 *Gut Springenheide and Tusky* [1998] ECLI:EU:C:1998:369, para 31; see also Council Directive 2005/29/EC concerning unfair business-to-consumer commercial practices [2005] OJ L149/22, recital 18.

49 See to that effect: Joined Cases C-240/98 to C-244/98 *Océano* [2000] ECLI:EU:C:2000:346; Loos (n 8) 188 f.

50 Mak (n 10) 59.

51 Loos speaks of the ‘average consumer targeted by the trader’s commercial activity’: M Loos, ‘Double Dutch – On the role of the transparency requirement with regard to the language in which standard contract terms for B2C-contracts must be drafted’ (2017) 6 *Journal of European Consumer and Market Law* 54, 56.

video game player that has already ‘participated in the market’ or wishes to do so.

b. Information to take into account

The average consumer has to take into account all information that surrounds the contract formation.⁵² This includes ‘promotional material and information provided by the (commercial party) in the negotiation’ of the contract.⁵³ In the great majority of contracts concluded online, such as in the gaming industry, the latter element is quasi-non-existent. The prevalence of contracts of adhesion simply does not allow consumers to enter contract negotiations. In regard to information provided by the commercial party, the CJEU clarified in *Lovaszé Tóth* how far the professional party has to go when informing the consumer. The case concerned a clause that needed to be read in conjunction with a Hungarian contract law provision.⁵⁴ The plaintiff, *Lovaszé Tóth*, argued that the company had to inform her about the meaning of that provision. The court disagreed, finding that it goes beyond the duties of the commercial party to educate the consumer about general contract law, provided that the clause itself was comprehensible (which it was).⁵⁵

Next to information provided actively during the conclusion of the contract, there is promotional material.⁵⁶ Promotional material is material that is easily accessible by the consumer, e.g. on the service’s website or a brochure that accompanies a contract form.⁵⁷ Whether information must be taken into account that can be found somewhere on the website hidden behind multiple hyperlinks is highly questionable. In *DAÄB*, a case concerning the ADR Directive, the CJEU ruled that all information required under that Directive must form part of the actual terms and conditions.⁵⁸ While not applicable to the UCTD, a similar approach might benefit the position of the consumer, provided the terms are transparent. Carefully looking through a website to find information that would clarify a certain term would put the consumer in an even weaker ‘position vis-à-vis the seller’. If the information cannot be found with a few clicks, it should not have legal effects.

c. Accessibility and presentation

In 1956, the English judge Lord Denning wrote:

*‘The more unreasonable a clause is, the greater the notice which must be given of it. Some clauses I have seen would need to be printed in red ink on the face of the document with a red hand pointing to it before the notice could be held to be sufficient.’*⁵⁹

While information displayed in red can actually be recalled less easily than information written in black,⁶⁰ Lord Denning made clear that the presentation of a contract plays an important role in assessing its content, thus linking formal and substantive transparency. Applying the red hand rule would be awfully impractical when concluding contracts online and luckily there are viable alternatives available.⁶¹

The prime example in contracting seems to be the use of uppercase letters to make the reader aware of a particular term. *Steam*’s Subscriber Agreement for instance features an all caps disclaimer at the beginning of the agreement. Terms that relate to *Steam*’s liability towards the consumer are always capitalised, too. Studies have shown that ‘all caps’ terms are less comprehensible than lower-case or standard writing, especially for non-native English speakers.⁶² While they might bring attention to a certain term, their inferior

level of comprehension means that they should be avoided. In another study, participants reading a text drafted in serif fonts recalled 9 % more information than participants that have read the text in other fonts.⁶³

Whether text should be drafted in boldface, italics, big or small text depends on the age of the reader. Studies have shown that high school students prefer smaller font sizes while second graders prefer a bigger size.⁶⁴ Bold and italicized text increases the difficulty of reading a text and therefore increases its disfluency.⁶⁵ Fluency is the ‘relative ease experienced during processing’.⁶⁶ Thus, in contrast, disfluency means that processing a text is difficult. Disfluency has been found to increase comprehensibility and memory of the text, since the reader needs to spend more time reading it.⁶⁷ However, while disfluency may be appreciable in the educational realm,⁶⁸ it – with no doubt – ‘reduces liking’ where the reader is not that motivated to read the text in the first place.⁶⁹ As most consumers are likely to be discouraged from reading terms and conditions because of their length and use of legalese,⁷⁰ relying on disfluent text to achieve comprehensibility does not seem to be a viable option.

There is ample evidence that capitalised, bold, or italic text increases disfluency and therefore discourages the reader from actually reading it.⁷¹ Additionally, studies suggest that when people know they are being asked questions they take considerably longer to read a text that is drafted in all caps compared to reading a normal text, while both groups have the same degree of comprehension.⁷² When used in moderation, such as for headings, emphasis could be useful. To test this empirically, differently formatted versions of the

52 Article 5 UCTD and the case law discussed previously.

53 *Árpád Kásler v OTP Jelzálogbank Zrt* (n 10) 74.

54 C-34/18 *Ottília Lovaszé Tóth v ERSTE Bank Hungary* [2019] ECLI:EU:C:2019:764, para 61.

55 *ibid* 70.

56 *Árpád Kásler v OTP Jelzálogbank Zrt* (n 10) 74.

57 See also: *Dellacasa* (n 11) 170.

58 C-380/19 *Verbraucherzentrale Bundesverband e. V. v Deutsche Apotheker- und Ärztebank e. G.* [2020] ECLI:EU:C:2020:498, para 29.

59 *J Spurling Ltd v Bradshaw* [1956] EWCA Civ 3, 1 WLR 461, 2 All ER 121, 1 Lloyd’s Rep 392.

60 J Beyon and C Cox-Boyd, ‘The Influence of Font Type and Color on Online Information Recall’ (2020) 22 North American Journal of Psychology 13-26.

61 Using a red hand as an image on a website, maybe even an animated one, reminds us more of a website from the early 2000s than anything else.

62 Y-T Kim and others, ‘Neural processing of lower- and upper-case text in second language learners of English: an fMRI study’ (2017) 33 Language, Cognition and Neuroscience 165, 169 ff.

63 M Gasser and others, ‘The Influence of Font Type on Information Recall’ (2005) 7 North American Journal of Psychology 181, 185.

64 T Katzir, S Hershko and V Halamish, ‘The Effect of Font Size on Reading Comprehension on Second and Fifth Grade Children: Bigger Is Not Always Better’ (2013) 8 PLoS One 1, 4 ff.

65 *ibid*.

66 I Hernandez and JL Preston, ‘Disfluency disrupts the confirmation bias’ (2013) 49 Journal of Experimental Social Psychology 178.

67 C Diemand-Yauman, DM Oppenheimer and EB Vaughan, ‘Fortune favors the bold (and Italicized): Effects of disfluency on educational outcomes’ (2011) 118 Cognition 111-115.

68 *ibid*.

69 R Reber, P Winkielman and N Schwarz, ‘Effects of Perceptual Fluency on Affective Judgements’ (1998) 9 Psychological Science 45.

70 JA Obar and A Oeldorf-Hirsch, ‘The biggest lie on the Internet: Ignoring the Privacy Policies and Terms of Service Policies of Social Networking Services’ (2018) 21 Information, Communication & Society.

71 Kim and others (n 63) 169 ff; S Greer, E Sowden and L Scharff, ‘Email format and instructions: Influences on reading times, retention, and preferences’ (*United States Air Force Academy*, 2004) <http://www.laurenscharrff.com/courseinfo/SL03/email_study.htm> accessed 29.6.2020.

72 Greer, Sowden and Scharff (n 71).

same text should be presented to different groups of participants.

In our study however, we have not taken these formal aspects into consideration and focussed entirely on the substantive transparency of terms and conditions, i.e. the possibility for a consumer to acknowledge and foresee her rights and duties under the contract.

d. Empirical research

A seemingly straightforward method to assess foreseeability is to conduct empirical research. Through empirical research, one could define the average consumer, assess whether the drafting of the terms misleads consumers, or test the terms substantively by asking participants to read them and describe their consequences. Some authors call for such empirical evidence to be used in consumer law court cases.⁷³ We wonder whether empirically tested readability formulas could help legal professionals answer this call, albeit in a more efficient, less time-consuming way. Our primary aim is to find a way for legal professionals to assess transparency, and thus, within EU context, foreseeability, without too much effort. Conducting an all-encompassing empirical study is timely, often costly, and many legal professionals are not trained in it.⁷⁴ Applying a readability formula constitutes a more workable approach.

e. Flesch's readability formula

Before discussing how readability formulas might help assessing foreseeability, a decision must be made in regard to the language we want to focus on. As the interpretation of foreseeability is – at least officially – left to the national courts,⁷⁵ methods of assessment should be created and evaluated for every individual language. Since this paper analyses terms commonly used in the gaming industry, the prevalent language in which most terms and conditions are drafted should be assessed. It is safe to assume that almost all internationally acting online service providers offer English-language versions of their terms and conditions. This is definitely the case in the gaming industry.⁷⁶ Depending on the size of the company, some might offer additional versions in German, French, or Spanish.⁷⁷ Documents drafted in languages spoken significantly less, such as Bulgarian, Dutch, or Czech, are usually offered by very large companies only. We consequently focus on English-language contracts. Since gaming terms and conditions are often not available in the consumer's native language, courts and supervisory bodies all across the globe need tools to objectively assess English language contracts. The probands of our study were native as well as non-native English speakers.

In 1948, Rudolf Flesch pioneered with the publication of his *Flesch Reading Ease* formula.⁷⁸ This formula aims at providing an objective way to analyse the readability of English-language texts. The formula outputs a reading level on a scale from 0 to 100.⁷⁹ A score of at least 60 is 'plain English', although a score of 80 or higher is recommended for conversational language. Therefore, the higher the score, the more readable the text. A later version of the formula, called *Flesch-Kincaid Grade Level* formula, shows how many years of US school education a reader would need in order to comprehend the text. Both versions are limited in that they cannot assess texts more complicated than what college graduates would understand.⁸⁰ While there are also other formulas to test readability,⁸¹ this paper focusses on the *Flesch Reading Ease* formula in its 1979 version since Flesch has actively explained and defended his formula for use in legal

documents after joining the US Federal Trade Commission (FTC) in the 70s.⁸² The two most important elements of Flesch's formula are word length in syllables and the average sentence length.⁸³ For the former, all syllables are counted and subsequently divided by the total number of words. The latter is calculated by dividing the total number of words by the total number of sentences. Both elements are then entered into a formula⁸⁴ that outputs the level of reading ease. The average US American adult understands texts with a score of roughly 65 to 80.⁸⁵

However, the effectiveness of readability formulas is disputed, and some studies doubt that such formulas have any inherent value since they can be tricked easily.⁸⁶ For example, drafters of legal texts could purposefully use very short sentences and words with few syllables, regardless of whether these words form part of the average consumer's vocabulary. Furthermore, readability formulas ignore aspects of formal transparency, such as ability to evaluate how the information is presented (typeface, graphic layout, headings) and how information is phrased (impersonality, nominalisations, idea-stuffed sentences, double negatives, illogical order). Readability formulas are also widely criticised for ignoring important aspects of readability such as the semantic and discourse levels of linguistic processing.⁸⁷ While there are attempts to create methods based on neuro-linguistic programming, these are not available to the public and feature a fundamentally higher level of complexity. However, readability formulas do allow an objective analysis of legal texts. This means that there is no human judgement involved. Additionally, while they were not designed to measure comprehensibility as such, a sufficient degree of plainness is necessary for a consumer to at least be able to comprehend a text. While transparency has multiple aspects, readability is a necessary feature thereof. Therefore, our hypothesis is that a 'readable' text can also be comprehended by the average consumer and that the formula can predict comprehensibility. In turn, this comprehensibility allows the consumer to foresee a term's con-

73 J Luzak, 'Empirical Evidence in Consumer Law Cases: What Are 'Up To' Claims Up To?' in B Heiderhoff and R Schulze (eds), *Verbraucherrecht und Verbraucherverhalten* (Nomos 2016); A Blackham, 'Legitimacy and empirical evidence in the UK courts' (2016) 25 Griffith Law Review 414-440.

74 M Malsch, 'Het recht is te belangrijk om aan juristen over te laten' (Inaugural lecture, Open Universiteit 2019) 7 f.

75 *Jean-Claude Van Hove v CNP Assurances SA* (n 36) 27 f.

76 The supplementary materials contain a list of the six industry leaders in the gaming industry, all offering their terms and conditions in English: <https://bit.ly/2ZofBhH>.

77 *ibid*.

78 R Flesch, 'A new readability yardstick' (1948) 32 Journal of Applied Psychology 221-233.

79 R Flesch, *The Art of Readable Writing* (Harper & Row 1949) 150.

80 TM Walsh and TA Volsko, 'Readability Assessment of Internet-Based Consumer Health Information' (2008) 53 Respiratory Care 1310, 1311.

81 For more information, see WH DuBay, 'The Classic Readability Studies' (*Impact Information*) <<https://files.eric.ed.gov/fulltext/ED506404.pdf>> accessed 29.6.2020 and GH McLaughlin, 'SMOG grading: A new readability formula' (1969) 12 Journal of Reading 639-646.

82 R Flesch, *How to Write Plain English: A Book for Lawyers and Consumers* (Harper & Row 1979). In this book, Flesch applies his formula to insurance policies, banking contracts, and more.

83 Flesch, *The Art of Readable Writing* (n 79) 213-216.

84 Flesch Reading Ease = $206,835 - 84,6 \times (\text{word length in syllables}) - 1,015 \times (\text{sentence length})$, see: Flesch, *How to Write Plain English: A Book for Lawyers and Consumers* (n 82) 24.

85 Walsh and Volsko (n 80) 1311.

86 KA Schriver, 'Plain Language in the US Gains Momentum: 1940-2015' (2017) 60 IEEE Transactions on Professional Communication 343, 351 ff; Van Boom, Desmet and Van Dam (n 18) 202 f.

87 SA Crossley and others, 'Predicting Text Comprehension, Processing, and Familiarity in Adult Readers: New Approaches to Readability Formulas' (2017) 54 Discourse Processes 340, 342 f.

sequences. Being able to foresee consequences is the most important aspect of the CJEU's current transparency test. In order to improve the results of a mere mathematical approach we have set out to develop a revised version of Flesch's formula that takes into account at least some of the mentioned criticisms.

f. An attempt at a revised formula

Flesch suggested changes to texts that seem rather obvious. Shorten them rigorously; delete information that is not strictly necessary for the reader. Use shorter sentences, replace polysyllabic words with 'easier' ones and avoid pre- and suffixes. These changes will increase readability if one was to trust Flesch's readability formula. However, as explained before, readability formulas such as Flesch's also have disadvantages. It is of course very difficult to translate findings of cognition and psychology into a simple formula. Nevertheless, it seems that certain criticisms could be addressed. We focus on one specific shortcoming of the formula, namely its disregard of specific meanings for specialised words, such as legalese.

Benson, another scholar active in this field, once described legalese as '*characterized by passive verbs, impersonality, nominalisations, long sentences, idea-stuffed sentences, difficult words, double negatives, illogical order, poor headings, and poor typeface and graphic layout*'.⁸⁸ Readability formulas only analyse certain of these aspects. They only look at the difficulty of the words by counting syllables and whether long sentences have been used. Relying on syllables per word to assess the difficulty of a word ignores the fact that terms more commonly used will be easier to understand regardless of their composition.⁸⁹ This aspect – namely the origin, meaning, and use of the words – is the only one that we address in a revised formula.

To take this factor into account, the output of the readability formula can be weighed against the amount of words that are used frequently. 'Oxford 3000' is a list compiled by Oxford University Press that encompasses the most 'frequently' used words across all disciplines.⁹⁰ The list is based on words that are crucial for communicating in English, although they are not most frequently used (such as body parts).⁹¹ It is therefore a list of crucial, while not very commonly used, words that is filled up with the most frequently used words in the English language, with a cap of 3000 words.

Let us illustrate the above by proposing these two sentences:

'You can delete all your information at the counter.'

'You can delete all your data at the counter.'

The first sentence scores a *Flesch Reading Ease* result of 66,1 (8th grade) while the second scores 84,9 (6th grade). Is the latter sentence really easier to read? The word 'information' is part of the Oxford 3000 list of common words, but it features four syllables. 'Data' on the other hand, a synonym for 'information', only has two syllables. It is also listed in the Oxford 3000. This first component of the argument against the use of syllables as sole indicator for unreadability therefore relates to the fact that words with many syllables may still be easy to read due to their frequent use in common language.

Now read these two sentences:

'She had a great goodbye after her visit.'

'She had a great congé after her visit.'

Example one scores 82,3 on Flesch's test and is thus readable by 6th graders. The second sentence on the other hand receives an 92,9: 'very easy' (5th grade). It would hardly make sense to assume that 5th graders, or even college graduates, know what a 'congé' is. It does not form part of the Oxford 3000 but 'goodbye' does. Additionally, words such as 'congé' also have specialised meanings (e.g. in law). And so do a lot of other words with few syllables. This is the second component of the argument: a word that is easy to read is not automatically easy to comprehend.⁹²

To address this issue, we have made changes to the readability formula by making use of the Oxford 3000 and a database of legal terms that is based on the 1910 version of Black's Law (*LexPredict*),⁹³ with the latter being extended by current day legalese. The *Flesch Reading Ease* relies, next to the average number of syllables per word, also on the average sentence length. It was therefore essential to not simply remove words from texts (because they are very common) or count some twice (for example in case of single syllable legal terms). Instead we opted for 'downgrading' words that are listed in the Oxford 3000 to single syllables, regardless of how many syllables they actually contain. The syllables of terms listed in Black's Law, on the other hand, were counted twice. In the eyes of this formula, common words are per se seen as easier to read while legal terms are per se harder. In the first example, the new formula led to a correction of the polysyllabic character of 'information'. Both sentences now reach 84,9. In the second example, the use of 'goodbye' now leads to a result of 92,9, therefore a higher readability, whereas the use of 'congé' leads to a harder to read 82,3.

We have decided to not include elements of text presentation in a revised formula. The inclusion of too many elements would have required a fundamentally bigger sample size. By focussing on substantive transparency, we are able to assess what readability formulas aim at, namely evaluating the readability and comprehensibility of pure, unformatted text.

III. Preliminary empirical evidence

To evaluate whether readability formulas can indicate if a term's consequences are foreseeable, and the term thus is transparent, we conducted an online study using a questionnaire. When analysing terms and conditions used by the six biggest digital distribution platforms in the gaming industry we found a number of terms that were commonly used by all of them.⁹⁴ We then opted to use excerpts from the actual terms and conditions used by one of these platforms: *Steam*. A large part of *Steam*'s terms and conditions has recently been evaluated by the High Court of Paris in its *UFC-Que Choisir v Valve* (the company behind *Steam*) decision.⁹⁵ Interestingly, the court paid little attention to the transpar-

88 RW Benson, 'The End of Legalese: The Game is Over' (1985) 13 *Review of Law & Social Change* 519, 531.

89 Seely (n 13) 120 ff.

90 'About the Oxford 3000 wordlist' (*Oxford Learner's Dictionaries*) <<https://www.oxfordlearnersdictionaries.com/about/oxford3000>> accessed 29.6.2020.

91 *ibid.*

92 Van Boom, Desmet and Van Dam (n 18) 202.

93 'LexPredict/lexpredict-legal-dictionary' (*LexPredict LLC*, 2017) <<https://github.com/LexPredict/lexpredict-legal-dictionary>> accessed 22.11.2018.

94 To see a comparison of the original and the excerpts used, please reference the supplementary materials: <https://bit.ly/2ZofBhH>.

95 *UFC Que choisir c. Valve Corp* (n 16).

ency principle and mainly focused on the unfairness test of the UCTD, instead.

This study has three purposes: first, to find out whether the traditional Flesch readability formula can be used to assess foreseeability, second, whether our revised version of that formula does a better job, and third, how foreseeable the consequences of terms and conditions in the gaming industry really are.

1. Methodology

Instead of presenting participants with original full-text versions of terms and conditions of different companies, we have chosen to rely on only one set of terms and conditions. *Steam's* terms and conditions were found to be most suitable for the following two reasons. First, *Steam* has been subject to both recent media and legal attention.⁹⁶ Second, as similar provisions can be found in terms and conditions of most of the other biggest companies in the gaming industry, too, they are representative for the gaming industry as a whole.

For the questionnaire the original text was first anonymised. All references made to *Steam* were replaced by references to the fictional GAMERZ platform. Second, we included parts of external documents the original terms only referred to by hyperlinks. We did this, even though we believe that consumers are not likely to read information placed on another website despite being declared integral parts of the terms and conditions, to also check the readability of these external documents. To keep the questionnaire short, participants were not confronted with the full text of the original terms and conditions. As a third step, the specific clauses used were chosen, which was a highly subjective decision.⁹⁷ Fourth, we opted to only assess the actual wording of the clauses and not their presentation or accessibility. Thus, we only tested substantive aspects of transparency.

Participants were recruited online via posts on platforms, such as reddit.com, Facebook.com and fok.nl. Also university facilities were used: we recruited students by sharing the link to the study on the online student environment and presenting it during lectures. University recruitment took place in law faculties only. Participants did not receive any compensation in exchange for participation.

We used Qualtrics to create an online questionnaire. Participants were asked to complete five general demographic questions about their gender, mother tongue, year of birth, highest level of completed education, and the types of online services used at least once per week. Participants were presented with a brief description of GAMERZ, the fictional company the terms and conditions were allocated to.⁹⁸ Once read, each participant was presented with three excerpts of GAMERZ's (i.e. *Steam's*) terms and conditions, one at a time.⁹⁹ Each excerpt was accompanied by one (multiple-choice) question about the consequences of the term. In most cases, the participants were able to choose between yes and no. Where more than two answers were provided, only one was correct (and thus counted towards the total number of correct answers). Correct answers were those answers that reflected the most likely, tendentially unanimous interpretation of the law. We assumed that a correct answer correlates with comprehensibility, which in turn allows the participant to foresee the excerpt's consequences. Furthermore, both the original *Flesch Reading Ease* formula and the revised formula were applied to each excerpt. This was done to assess whether a high readability score also corresponded to a high

number of correct answers and thus comprehensibility and foreseeability.

We prepared a total of eight different excerpts, each featuring different readability scores on both formulas. Each participant was presented with three of these excerpts. The excerpts were selected at random. Consequently, not every participant was given the same excerpts and questions.¹⁰⁰ Once all questions had been answered, participants were thanked and given our contact information.

We sourced information on the current average gamer in the EU from existing studies. In 2018, the largest groups were gamers aged 11-14 (84 % of which played video games), 6-10 (76 %), and 15-24 (74 %),¹⁰¹ while the fastest growing age group was 25-34 year olds.¹⁰² On average, the European gamer is 31 years old.¹⁰³ Of all EU gamers, 46 % are women.¹⁰⁴ There are no data available on the average education level of gamers. In the US, more than 86 % of all gamers have attended at least one year of college.¹⁰⁵ Unfortunately, these data do not differentiate between daily, weekly, monthly, or even yearly gamers. Thus even people that only play once a year are also included.

Using this information, we tested the following hypotheses. First we assumed there would be no gender difference as far as comprehension is concerned. Second, native English speakers would score higher on comprehensibility compared to non-native speakers. This was based on the assumption that native English speakers are more likely to comprehend English legalese because they have encountered English legal terms more frequently than non-native speakers. Third, adolescents and young adults (ages 12-25 years) would score higher on comprehensibility compared to older participants because they spend more time using internet services than other age groups and therefore have been confronted with a larger number of similar terms in the past.¹⁰⁶ Fourth, we expected that individuals with higher levels of completed education would have higher levels of comprehension. Fifth, participants using a higher number of different online services, such as Facebook, at least once per week, would have higher comprehension. Sixth, and last, we assumed that a term scoring higher on the revised formula than on the original formula would be more comprehensible and inversely, a term scoring lower on the revised formula than on the original formula would be less comprehensible than the latter formula makes us think.

96 S Sundier, 'France court says digital games can be resold' (*Jurist.org*, 20.9.2019) <<https://www.jurist.org/news/2019/09/france-court-says-digital-games-can-be-resold/>> accessed 29.6.2020.

97 To see a comparison of the original and the excerpts used, please reference the supplementary materials: <https://bit.ly/2ZofBhH>.

98 *ibid*.

99 The supplementary materials contain images of the online questionnaire: <https://bit.ly/2ZofBhH>.

100 Some participants were recruited during a lecture and could have completed the questionnaire next to one another. In order to ensure that participants could not copy one another from visible screens, questions were randomised.

101 ISFE, 'Key Facts 2019' (*Interactive Software Federation of Europe*) <<https://www.isfe.eu/wp-content/uploads/2019/08/ISFE-Key-Facts-Brochure-FINAL.pdf>> accessed 29.6.2020, 6.

102 *ibid* 7.

103 *ibid*.

104 *ibid* 8.

105 C Gough, 'Distribution of video gamers in the United States as of January 2018, by education level' (*statista*, 30.10.2018) <<https://www.statista.com/statistics/499707/share-consumers-ever-play-video-games-by-education-level-usa/>> accessed 29.6.2020.

106 J Clement, 'Average duration of daily internet usage worldwide' (*statista*, 2019) <<https://www.statista.com/statistics/416850/average-duration-of-internet-use-age-device/>> accessed 29.6.2020.

2. Results

The initial sample included 168 participants. 96 participants (57,1 %) completed the full questionnaire. Participants who did not complete the questionnaire were excluded from the subsequent analyses. There were equal numbers of male and female participants, and their approximate ages ranged from 17 to 62 years. See Table 1 for descriptive information on participants age, gender, language, education, and service use.

Table 1. Sample demographics

Descriptor	N = 96
Age <i>M</i> (<i>SD</i>) ^a	25,81 (10,19)
Gender (<i>n</i>)	
Female	47
Male	47
Rather not say	2
Language (%) ^b	
English	26,0
Dutch	25,0
Other	47,9
Missing	1,1
Education (%)	
High school graduates	45,8
Vocational training	2,1
Bachelor's degree	21,8
Master's degree	20,8
Doctoral degree	3,1
Number services used <i>M</i> (<i>SD</i>)	3,0 (1,14)
Service used (%) ^c	
Music streaming	69,8
Movie/TV streaming	68,8
Social media	96,9
Gaming	31,3
Shopping	31,3

a Approximate age of participant (2019 – reported year of birth).

b One participant reported speaking both English and Dutch, counted in English category.

c Percentage of participants who used a service in this category once per week.

On average each of the eight excerpts was read by 36 participants (the real number ranging from 29 to 41). We calculated the average readability (both with the original and revised formula) for the three presented excerpts per participant. The average excerpt readability that participants were presented with was 36,4 (*SD* = 5,7) on the original scale and 34,6 (*SD* = 7,5) on the revised scale. This corresponds to a hard to understand text for college attendees (30-50),¹⁰⁷ reaching almost college graduate difficulty (0-30). By chance, males were presented with excerpts with a slightly lower average readability (original/revised readability score: *M* = 35,3/33,2, *SD* = 6,03/8) than females (*M* = 37,7/36,4, *SD* = 5,07/6,7; original *t*(92) = 2,17, *p* = 0,033; revised *t*(92) = 2,11, *p* = 0,037).

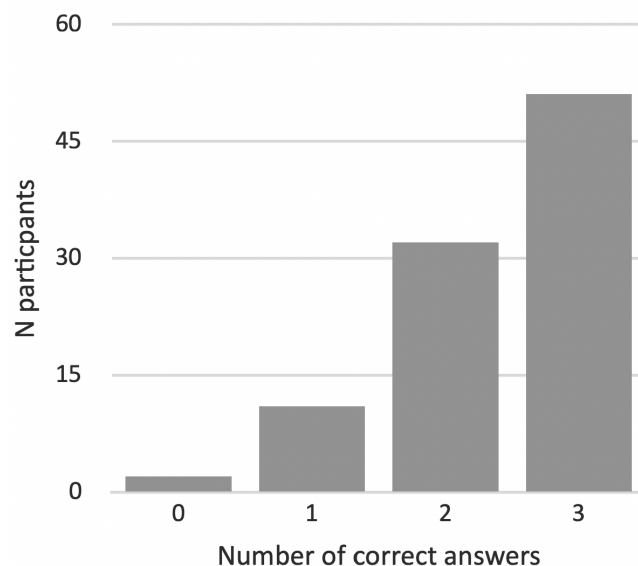


Figure 1. Number of participants who answered 0-3 questions correctly.

On average, participants answered 2,38 items correctly (see Figure 1). For example, a huge majority (81,4 %) were able to understand that, as consumers being situated in Germany, they do not have to bring a lawsuit with a court in Washington, USA. Many participants understood that they would lose access to all their games if they violated the code of conduct (75,8 %) or any other term (91,4 %). 75,6 % realised that they would lose access to their games if they decided to disagree with a unilateral change of terms and conditions. Additionally, 85,4 % of all participants foresaw that they would not be allowed to resell their games. Each item was answered correctly 78,78 % of the time. There were no differences between males and females regarding the number of correct answers (*t*(92) = -0,94, *p* = 0,348). There were also no differences regarding the number of correct answers between native and non-native English speakers (*t*(93) = -0,063, *p* = 0,95). Additionally, there was no significant correlation between the number of online services used and the total number of correct answers (*r* = -0,004, *p* = 0,97). Adolescents and young adults made up 68,8 % of the sample, but they did not use more online services than older participants (*t*(93) = -0,131, *p* = 0,896). The level of completed education had no effect on the number of correct answers (*F* (4,91) = 0,443, *p* = 0,777).

Figure 2 shows the most readable to least readable excerpts based on the original (blue) and revised (orange) formula and the correlating number of correct answers. For example, the most readable excerpt on the original formula was understood by around 75 % of participants, while the most readable excerpt on the revised formula was understood by less than 70 %.

It shows that there is no correlation between the readability as assessed by *either* formula and the comprehensibility of the terms (number of correct answers).

¹⁰⁷ Flesch, *How to Write Plain English: A Book for Lawyers and Consumers* (n 82) 26.

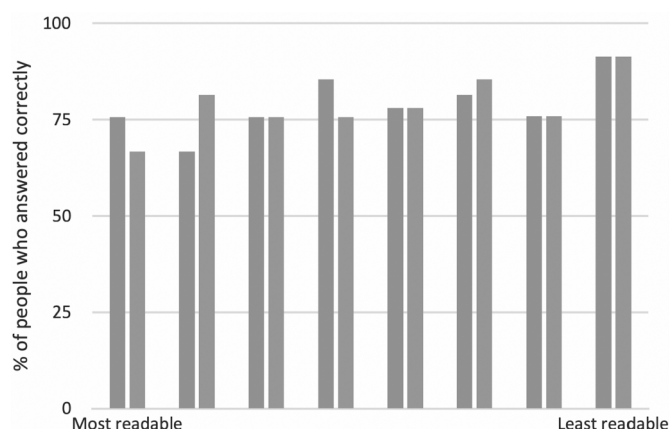


Figure 2. Most readable to least readable excerpts on the Flesch formula (left) and the revised formula (right) with the percentage of corresponding correct answers.

3. Discussion and limitations

This study was conducted 1) to evaluate whether there is an empirical basis for readability formulas (including our revised formula) to be used to assess the comprehensibility of terms, and consequently their foreseeability (see methodology), and 2) to assess the comprehensibility of common gaming terms and conditions. To depart from the recurring focus on terms and conditions regarding the financial services sector found in case law and literature, we chose common terms used in the gaming sector for our case study.

Results confirmed our hypothesis that there are no differences between males and females in comprehending gaming terms and conditions. There was no difference regarding the average number of correct answers between males and females. We expected that native English speakers would have an obvious advantage over non-native English speakers when reading and comprehending legal texts. Interestingly, this was not the case. A little more than one fourth of the sample were native English speakers, but there were no indications that this influenced comprehension either negatively or positively. We also expected that adolescents and young adults would score higher on comprehensibility than older participants. This was based on the assumption that younger participants would spend more time using a greater number of online services (and therefore, have more frequent exposure to the terms and conditions of those services). Contrary to our hypothesis, adolescents/young adults and older participants did not differ on the average number of correct answers. This may be, in part, due to the finding that the two groups on average used the same number of online services. Additionally, we did not find a difference in the use of gaming services between age groups. Existing studies suggest that the average European gamer is 31, and thus not an adolescent or young adult.¹⁰⁸ However, our results show that neither is the case: every age group features the same number of gamers. We also expected to see differences between levels of completed education on the number of correct answers. We found no significant differences on the average number of correct answers between the four education level groups.

Last but not least, our hypothesis that the revised formula will be a better indicator for comprehensibility was clearly rejected. As Figure 2 shows, the most readable excerpts on the revised scale (orange) did not receive more correct

answers than the least readable on the original scale. To the contrary, the least readable excerpt on both the original and revised scale received the highest number of correct answers!

As seen above, most of our hypotheses have been proven wrong. This offers important insights into current day contracting and allows us to draw the following conclusions based on this preliminary evidence. First, there was no correlation between an excerpt's readability score and the number of correct answers, providing preliminary evidence that the *Flesch Reading Ease* readability formula may not be a good indicator of comprehensibility of gaming terms and conditions. Second, our revised formula did not perform any better than the original. This is interesting to see as many of Flesch's critics complained about the use of syllable and word counts as the formula's only assessment methods. However, even taking one more element into account – namely the origin, meaning, and use of the words – did not improve the correlation with comprehensibility. Third, our data showed that the terms and conditions presented were widely understood by participants. 86,4 % of the sample answered at least two questions correctly, 53,1 % got all the answers right. Fourth, terms and conditions drafted in English were equally understood by native and non-native English speakers. This might confirm preliminary evidence that playing a lot of games leads to a better command of English.¹⁰⁹

As far as the comprehensibility of the terms is concerned, we have therefore reached two unexpected conclusions. It appears that the consequences of the terms were largely foreseeable, i.e. (substantively) transparent, and that (poor) readability – based on the formulas – is not a good indicator for (a lack of) comprehensibility. It is therefore likely that our assumption that readability largely equates to comprehensibility, and thus foreseeability, is wrong.

Several limitations should, however, be considered when interpreting these results. First, the current study relied on a small number of participants (96), which may not be a sufficiently large group to draw clear conclusions. Future research should aim at a larger sample size.

Second, the sample might not have been representative, as it was not selected to reflect the average gamer. Offline recruitment was only conducted at universities in the Netherlands. Dutch nationals tend to score high on English proficiency tests,¹¹⁰ and foreign students mainly follow English-taught programmes. Most importantly, the participants who were recruited offline were law students/professionals, only. Although we primarily targeted first-year students expecting them to have little to no knowledge of the law, they – by choice of their study – prove to at least have an interest in legal topics and thus in terms and conditions and licensing agreements. Participants recruited online could have been of all ages and educational backgrounds.

108 ISFE, 7; In the US, the average age is 33: entertainment software association, '2019 Essential Facts About the Computer and Video Game Industry' (esa, 2019) <https://www.theesa.com/wp-content/uploads/2019/05/ESA_Essential_facts_2019_final.pdf> accessed 29.6.2020.

109 KS Horowitz, 'Video Games and English as a Second Language' (2019) 11 American Journal of Play 379-410; P Sunqvist and LK Sylvén, 'Language-related computer use: Focus on young L2 English learners in Sweden*' (2014) 26 ReCALL 3-20; LT Tuan, 'Vocabulary Recollection through Games' (2012) 2 Theory and Practice in Language Studies 257-264.

110 Education First, 's Werelds grootste ranglijst van landen en regio's op basis van Engelse vaardigheden' (EF EPI, 2018) <<https://www.ef.nl/epi/>> accessed 29.6.2020.

That most participants were highly educated may not be a problem in terms of representativeness. Although there is no data on this for the EU, US data indicate that gamers generally enjoy a high level of education.¹¹¹ The mean (approximate) age of participants was however only 25,81 years old (instead of 31). Future research should properly represent the average consumer of videogames.

Third, participants were given short excerpts which they had to respond to directly. We thus essentially separated formal transparency from substantive transparency. In a real-life scenario, this would not be possible. If participants were required to collect and connect information found on different pages and/or documents, outcomes could vary. Although not directly concerned with comprehensibility and thus foreseeability, future research should also present terms using different methods, such as printed on paper, projected on a screen, or deeply nested within a website.

IV. Closing remarks

Readability formulas have received plenty of praise from legal professionals.¹¹² Drafters of terms and conditions like to refer to readability formulas as proof that their texts are written in ‘easy language’. Evidence of judges using them, however, is scarce and before spurring judges to apply readability formulas, we wanted to empirically assess whether these formulas are able to test transparency. Legal professionals have so far used these formulas to test the consumer friendliness of texts, even though there is no empirical proof that they actually work.

The benefits of readability formulas are clear: they can be applied by means of computers and are immune to subjectivity. Critics, however, second-guess the value of this objectivity. According to them more syllables are not in direct relation to being harder to read. Moreover, every word can have more than one meaning, some meanings might be very specialised depending on the context. In an attempt to (partly) set aside this criticism, we created a more refined formula based on *Rudolf Flesch’s Reading Ease* formula. This looked promising at first as texts containing legal terms were given a lower readability score than texts which contained ‘simple’ words found in the Oxford 3000. When compared to the original *Flesch Reading Ease* formula, our version allowed for a better reflection of the use of legalese.

When we tested both formulas empirically, however, our initial enthusiasm was dampened. In fact, neither the *Flesch Reading Ease* formula nor our revised legalese formula could predict whether a term’s consequences were comprehensible, and thus foreseeable. There is no correlation between the

output of either formula and the amount of correct answers given by our participants. Consequently, both readability formulas should be discarded as not being able to evaluate foreseeability. This does not mean that we found that these formulas never work, they simply do not work to assess transparency, as defined by the CJEU, of terms and conditions in the gaming industry. It may also be that the formulas succeed in measuring readability, but that our assumption that readability equates to comprehensibility, and thus foreseeability, is wrong.

Even though the results of our main research question indicate that readability formulas, including ours, may not be suitable to measure foreseeability and thus cannot predict transparency, there are important lessons to be learnt.

Next to this main finding, our study led to two other interesting findings. First, legal scholars, us included, are quick to assume that consumer contracts must be written in the native language of the consumer. However, our results indicate that English could be used as a universal language for terms and conditions in the gaming industry, as comprehensibility scores did not differ based on English native status. As the sample was highly educated, further research across different educational groups would definitely be needed.

Second, the results indicate that the consequences of gaming terms and conditions can be understood, and thus foreseen. This is good news. The terms we presented to our participants were transparent in the substantive sense of transparency. However, aspects of formal transparency, such as the accessibility of the terms have not been assessed. The main issue with transparency may well lie with those formal aspects. This is an important preliminary insight for the courts: when assessing transparency of terms and conditions, judges should pay attention to aspects of formal transparency and not only focus on the foreseeability requirement and substantive transparency in the strict sense. The recent *Profi Credit Polska*-judgment is a step in the right direction, as the CJEU considered relevant ‘whether the consumer has actually been given the opportunity to examine (the terms’) content’.¹¹³ Indeed, as follows from our study, if the consumer accesses and reads the terms, she might well understand them. ■

¹¹¹ Gough (n 106).

¹¹² Indirect praise can be found in the large amount of readability legislation introduced especially in the United States. For example, both Florida and Connecticut require insurance policies to score a minimum of 45 on the Flesch Reading Ease scale: § 627.4145 Fla. Stat. (2011); CT. Gen. Stat. § 38a-295 (2013).

¹¹³ C-186/16 *Joined cases C-419/18 and C-483/18 Profi Credit Polska* [2019] ECLI:EU:C:2019:930, para 58.

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